

Experiment-7

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Course Code: -MLA0305

Slot-B

Title: - Temporal Difference Learning Using TD (0) Algorithm

Aim- To implement the Temporal Difference TD (0) learning algorithm for estimating the value function directly from experience without waiting until the end of an episode.

Procedure

1. Initialize the environment and value function.
2. Select an initial state.
3. Select an action according to the policy.
4. Execute the action in the environment.
5. Observe the next state and reward.
6. Calculate the TD target.
7. Calculate the TD error using:
$$\delta t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$
8. Update the value function using:
$$V(S_t) \leftarrow V(S_t) + \alpha \delta t$$
9. Continue the process until the episode terminates.
10. Repeat for multiple episodes.
11. Plot the value estimates or learning performance.
12. Analyze the convergence of the value function.

Source Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
from IPython.display import display, FileLink
from google.colab import files
np.random.seed(42)
print("=" * 80)
print("EXPERIMENT 7")
print("TEMPORAL DIFFERENCE LEARNING USING TD(0)")
print("=" * 80)
N = 500
base = np.column_stack([
    np.random.randint(0, 10, N),
    np.random.randint(0, 10, N),
    np.random.uniform(-5, 10, N),
    np.random.uniform(0.85, 0.99, N)
])
```

```

gmm = GaussianMixture(
    n_components=3,
    random_state=42
)
gmm.fit(base)
generated = gmm.sample(N)[0]
dataset = pd.DataFrame(
    generated,
    columns=[
        "State",
        "Next_State",
        "Reward",
        "Gamma"
    ]
)
dataset["State"] = dataset["State"].clip(0, 9).round().astype(int)
dataset["Next_State"] = dataset["Next_State"].clip(0,
9).round().astype(int)
dataset["Reward"] = dataset["Reward"].clip(-5, 10).round(2)
dataset["Gamma"] = dataset["Gamma"].clip(0.85, 0.99).round(3)
print("\n10 SAMPLE DATASET RECORDS")
display(dataset.head(10))
print("Total Records:", len(dataset))
display(
    dataset.describe().T[
        ["min", "max", "mean", "std"]
    ].round(3)
)
filename = "Experiment_7_TD0_GMM_Dataset.csv"
dataset.to_csv(
    filename,
    index=False
)
print("\nDOWNLOAD DATASET:")
display(FileLink(filename))
files.download(filename)
N_STATES = 10
EPISODES = 500
ALPHA = 0.10
GAMMA = 0.90
def generate_episode():
    state = 0
    episode = []
    while state != 9:

```

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        next_state = min(
            state + 1,
            9
        )
        reward = (
            10
            if next_state == 9
            else -1
        )
        episode.append(
            (
                state,
                reward,
                next_state
            )
        )

        state = next_state

    return episode

# =====
# 7. TD(0)
# =====

V_td = np.zeros(
    N_STATES
)

td_history = []
td_errors = []

for episode in range(EPISODES):

    ep = generate_episode()

    for state, reward, next_state in ep:

        target = (
            reward +
            GAMMA *
            V_td[next_state]
        )

        error = (

```

```

        target -
        V_td[state]
    )

    V_td[state] += (
        ALPHA * error
    )

    td_errors.append(
        abs(error)
    )

    td_history.append(
        np.mean(V_td)
    )

# =====
# 8. MONTE CARLO COMPARISON
# =====

V_mc = np.zeros(
    N_STATES
)

returns_sum = np.zeros(
    N_STATES
)

returns_count = np.zeros(
    N_STATES
)

for _ in range(EPISODES):

    ep = generate_episode()

    G = 0
    visited = set()

    for state, reward, next_state in reversed(ep):

        G = reward + GAMMA * G

        if state not in visited:

```

```

        returns_sum[state] += G
        returns_count[state] += 1

    V_mc[state] = (
        returns_sum[state]
        /
        returns_count[state]
    )

    visited.add(state)

# =====
# 9. COMPARISON TABLE
# =====

comparison = pd.DataFrame({
    "State": range(N_STATES),
    "TD_0_Value": V_td,
    "Monte_Carlo_Value": V_mc,
    "Absolute_Difference": np.abs(V_td - V_mc)
}).round(3)

print("\nTD(0) VS MONTE CARLO")

display(comparison)

# =====
# 10. SUMMARY
# =====

summary = pd.DataFrame({
    "Metric": [
        "Dataset Records",
        "Episodes",
        "States",
        "Learning Rate Alpha",
        "Discount Gamma",
        "Average TD(0) Value",
        "Average MC Value",
        "Mean Absolute Difference",
        "Final TD Error"
    ],
    "Result": [
        len(dataset),

```

```

        EPISODES,
        N_STATES,
        ALPHA,
        GAMMA,
        round(np.mean(V_td), 3),
        round(np.mean(V_mc), 3),
        round(np.mean(np.abs(V_td - V_mc)), 3),
        round(td_errors[-1], 4)
    ]
})

print("\nSUMMARY RESULTS")

display(summary)

# =====
# 11. VISUALIZATION 1
# REWARD DISTRIBUTION
# =====

plt.figure(figsize=(10, 6))

plt.hist(
    dataset["Reward"],
    bins=25,
    color="#339af0",
    edgecolor="black"
)

plt.title(
    "GMM Synthetic Dataset - Reward Distribution",
    fontsize=16,
    fontweight="bold"
)

plt.xlabel("Reward")
plt.ylabel("Frequency")

plt.show()

# =====
# 12. VISUALIZATION 2
# TD LEARNING CURVE
# =====

```

```
plt.figure(figsize=(11, 6))

plt.plot(
    td_history,
    color="#20c997",
    linewidth=3
)

plt.title(
    "TD(0) Learning Progress",
    fontsize=16,
    fontweight="bold"
)

plt.xlabel("Episode")
plt.ylabel("Average State Value")

plt.grid(alpha=0.25)

plt.show()

# =====
# 13. VISUALIZATION 3
# TD ERROR
# =====

plt.figure(figsize=(11, 6))

plt.plot(
    td_errors,
    color="#e03131",
    alpha=0.55
)

plt.title(
    "TD(0) Prediction Error",
    fontsize=16,
    fontweight="bold"
)

plt.xlabel("Update")
plt.ylabel("Absolute TD Error")

plt.grid(alpha=0.25)
```

```

plt.show()

# =====
# 14. VISUALIZATION 4
# TD VS MC
# =====

plt.figure(figsize=(11, 6))

plt.plot(
    range(N_STATES),
    V_td,
    marker="o",
    linewidth=3,
    color="#7950f2",
    label="TD(0) "
)

plt.plot(
    range(N_STATES),
    V_mc,
    marker="s",
    linewidth=3,
    color="#ff922b",
    label="Monte Carlo"
)

plt.title(
    "TD(0) vs Monte Carlo Value Estimates",
    fontsize=16,
    fontweight="bold"
)

plt.xlabel("State")
plt.ylabel("Value")

plt.legend()
plt.grid(alpha=0.25)

plt.show()

# =====
# 15. VISUALIZATION 5
# ABSOLUTE DIFFERENCE
# =====

```



```
plt.figure(figsize=(11, 6))

plt.bar(
    range(N_STATES),
    np.abs(V_td - V_mc),
    color="#f06595",
    edgecolor="black"
)

plt.title(
    "Absolute Difference Between TD(0) and Monte Carlo",
    fontsize=16,
    fontweight="bold"
)

plt.xlabel("State")
plt.ylabel("Absolute Difference")

plt.show()

print("\nExperiment 7 completed successfully.")
```

10 SAMPLE DATASET RECORDS

	State	Next_State	Reward	Gamma
0	6	7	1.47	0.925
1	8	4	7.33	0.850
2	6	1	9.49	0.933
3	9	4	10.00	0.916
4	6	6	8.85	0.920
5	6	1	6.41	0.907
6	7	0	2.09	0.873
7	7	8	7.22	0.850
8	8	1	7.01	0.928
9	5	4	6.54	0.885

FIG: - 1 SAMPLE DATA

	min	max	mean	std
--	-----	-----	------	-----

State	0.00	9.00	4.348	2.721
Next_State	0.00	9.00	4.338	2.625
Reward	-5.00	10.00	2.558	4.349
Gamma	0.85	0.99	0.921	0.036

FIG: - 2 MEAN & SD

	State	TD_0_Value	Monte_Carlo_Value	Absolute_Difference
0	0	-1.391	-1.391	0.0
1	1	-0.434	-0.434	0.0
2	2	0.629	0.629	0.0
3	3	1.810	1.810	0.0
4	4	3.122	3.122	0.0
5	5	4.580	4.580	0.0
6	6	6.200	6.200	0.0
7	7	8.000	8.000	0.0
8	8	10.000	10.000	0.0
9	9	0.000	0.000	0.0

FIG: - 3 SUMMARY RESULTS

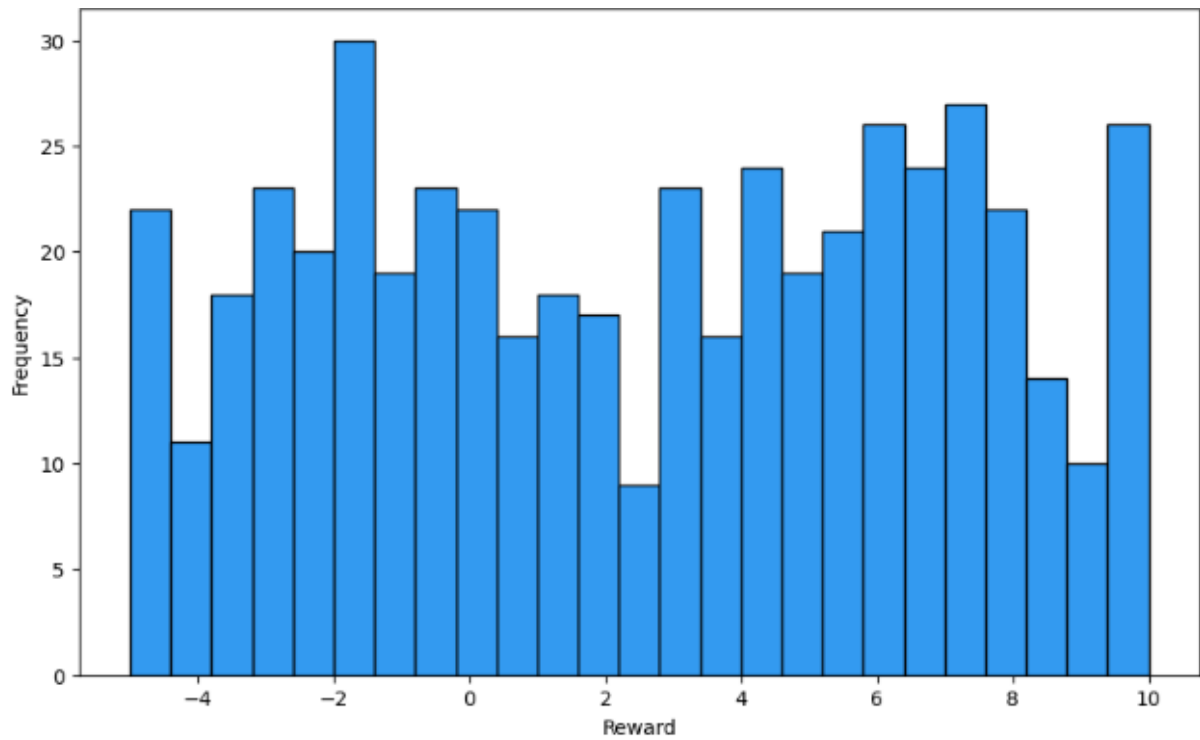


FIG: - 4 DATA SET REWARD DISTRIBUTION

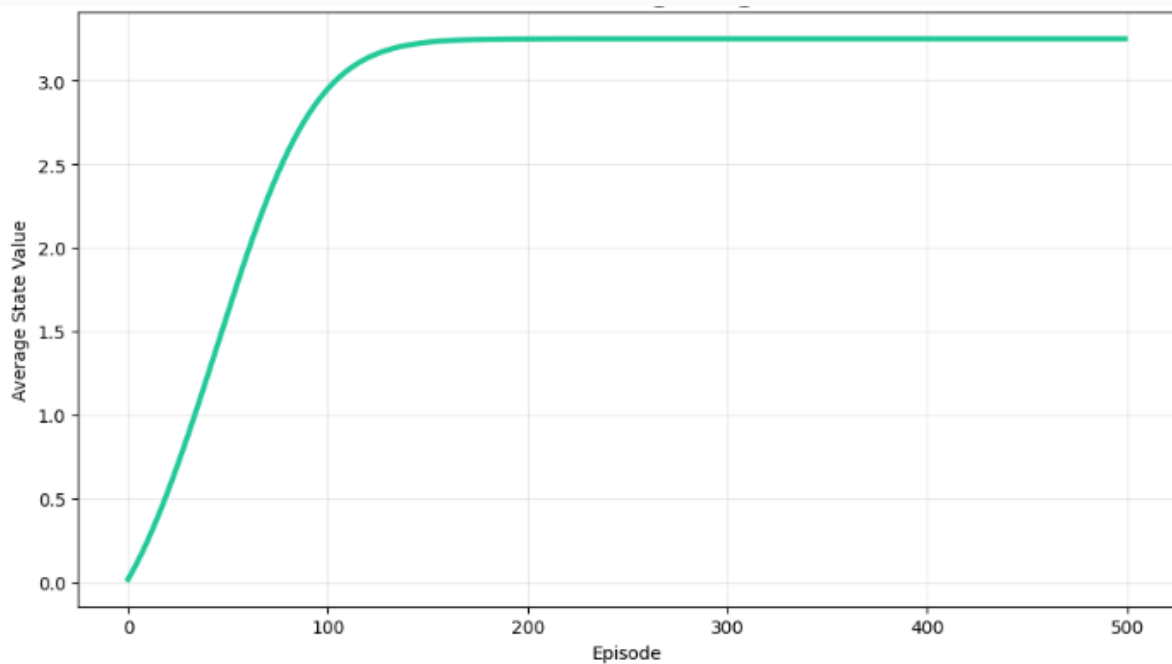


FIG: - 5 TD (0) LEARNING PROGRESS

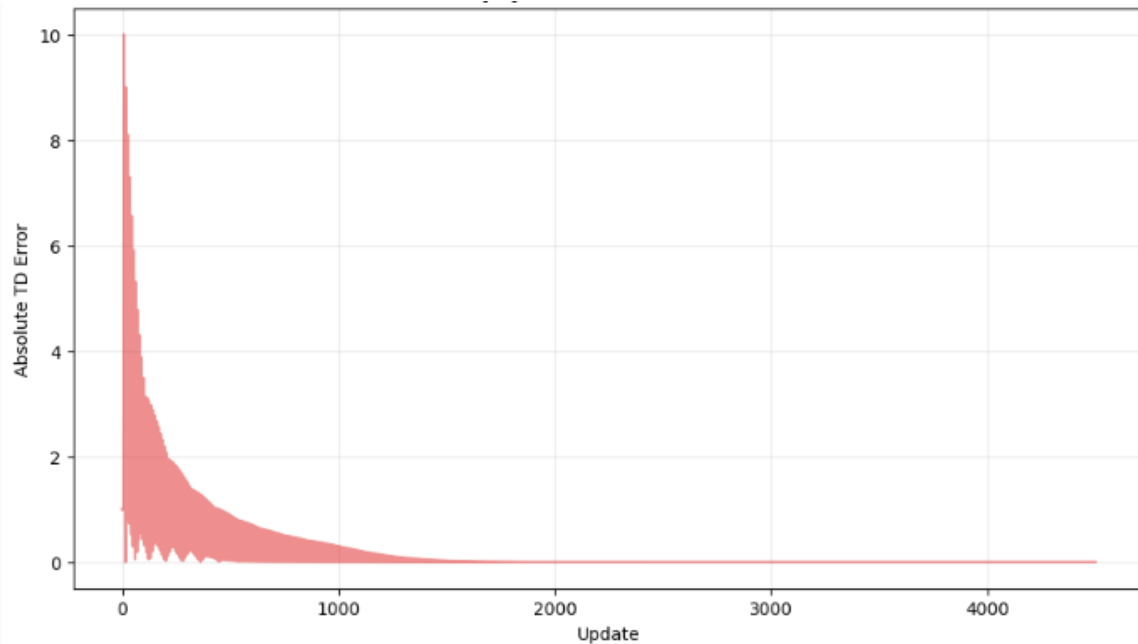


FIG: - 6 TD (0) PREDICTION ERROR

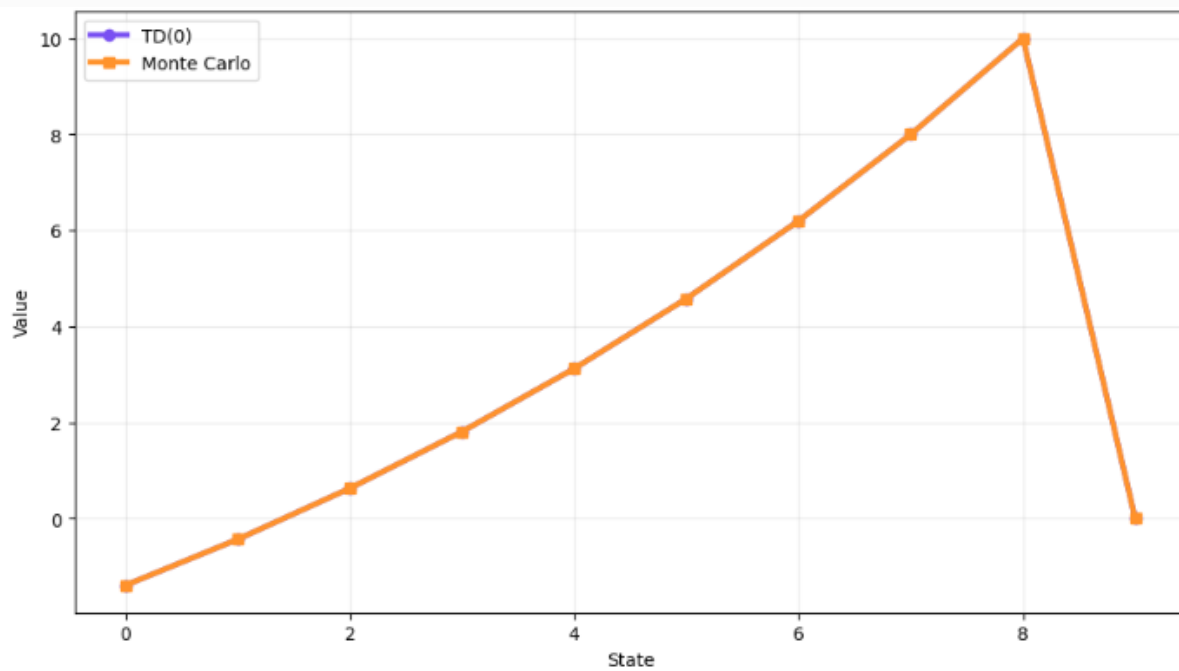


FIG: - 7 VALUE ESTIMATION

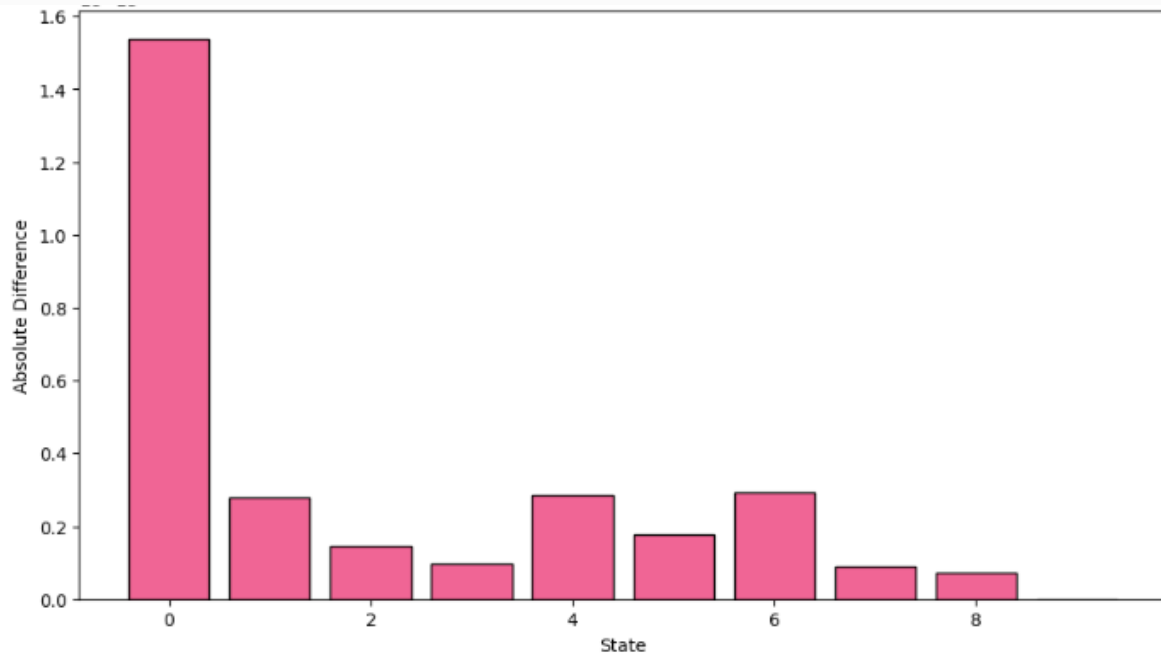


FIG: - 8 DIFFERENCE BETWEEN MC AND TD (0)

Result

Monte Carlo Prediction and Control were successfully implemented. The agent learned from complete episodes and gradually improved its policy based on the estimated returns.